THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM THREE
041 BASIC MATHEMATICS

Time: 3 Hours **Wednesday, 26th August 2026 a.m.**

Instructions

1. This paper consists of sections A and section B with a total of 14 questions.
2. Attempt all questions in both sections.
3. Each question in section A carries 6 marks while each question in section B carries 10 marks.
4. Mathematical tables and non-programmable calculator may be used.
5. Write your examination number on every page of your answer sheets.

SECTION A (60 MARKS)

1. (a) Find the sum of the LCM and GCF of 15,18 and 21
   (b) Solve the following equation
   $\frac{y}{15} = 2.\overline{45}$

2. (a) Express $\frac{1+tan60^0}{1-tan60^0}$ in surd form and rationalize the denominator.
   (b) Solve Log $x=\frac{Logx^2-4x-8}{Log100}$

3. In a survey of 100 randomly villagers, a marketing questionnaires included the following questions:

   1. Do you own a TV?
   2. Do you own a Car?
   3. Do you own a TV and a Car?

   75 answered yes to 1, 45 answered yes to 2, and 35 answered yes to 3

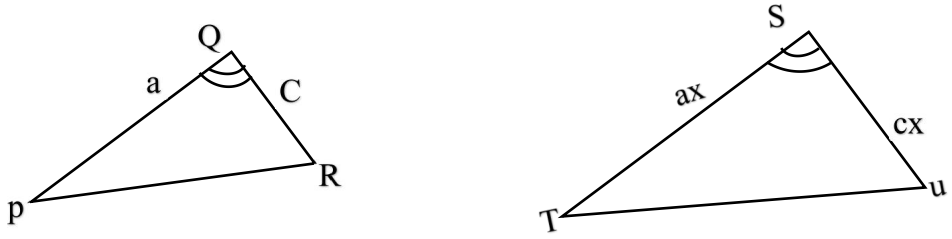
   i. How many villagers owned either a car or a TV?
   ii. How many did not own either a car or a TV?

4. (a) If $y\ mx+k$ is an equation of a straight line passing through the point $(-4,5)$ *and* $(-2,1)$, find the value of m and k.

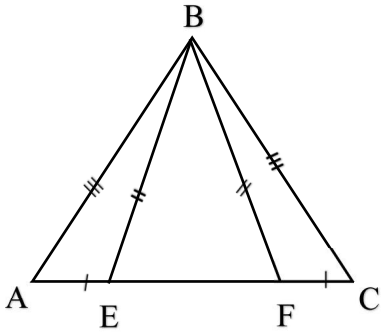
   (b) Find the equation of a line whose $y-$intercept is the same as that of a line $4x+y+6=0$ and the product of their gradient is -1.

5. (a) The following triangles are similar
   (i) Write a similarity statement

(ii)Give a mathematical reasoning that the two triangles are similar



(b) Given that $\overline{AE} \cong \overline{FC}$, $\overline{AB} \cong \overline{BC}$, and $\overline{BE} \cong \overline{BF}$. Prove that $\Delta AFB \cong \Delta CEB$



6. (a) Seypalm bought seedling trees to be planted on each side of his house road. The length of the road is 650 m long and the distance between one seedling tree and another is 13 m. How much cost him if the cost of one seedling tree is 3 500 shillings.

(b) If y varies directly as x and inversely as z, and if $y=10$ when $x=8$ and $z=5$, find the following:
   i. law connecting x , y, *and* z
   ii. y when $x=6$ *and* $z=2.5$

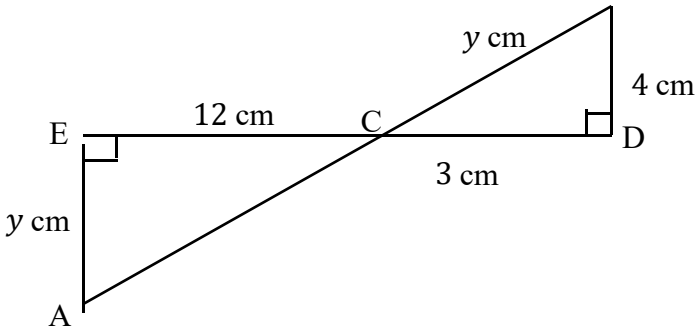
7. (a) David , Amina , and Nuru shared some oranges. The ratio of the number of oranges received by David and Amina was 7:3. The number of oranges David received to that of Nuru was in the ratio 4:5. If Nuru received 21 more oranges than David, find the number of oranges that Amina received.

   (b) The sides of a triangle are in the ratio of 6: 8: 10. If the perimeter of the triangle is 288 cm, find the difference between the longest side and the shortest side.

8. (a)The sum of the 3rd term and 2nd term of an arithmetic progression (A.P) is 19. If the 5th term exceeds twice the 2nd term by 8, find the sum of the first 50 terms

   (b) Find the 8th term of the following geometric progression sequence: $\frac{1}{64}, \frac{-1}{32}, \frac{1}{16}$, …

9. (a) How long will a person walk from point A to B in the following figure:



(b) If Sin $\theta = \frac{12}{13}$, find the value of Cos θ and tan θ without using right angled triangle.

10. (a) i. Use the concept of difference of two squares to find the exact value of the following $17\,972^2 - 17\,872 \times 17\,872$.

ii. Solve for h : $S = \pi r\sqrt{r^2 + h^2}$.

(b) Solve : $x^2 - 19x - 372 = 0$ by using the factorization method.

SECTION B (40 MARKS)

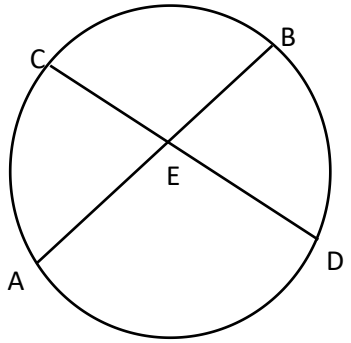
11. (a) The following table shows the masses of 100 students to the nearest kg.

Mass (kg)	40-42	43-45	46-48	49-51	52-54	55-57
Frequency	5	20	x	25	10	5

(i) Calculate the mode.

(ii) Estimate the median masses by using a cumulative frequency curve.

(b) In the following figure $\overline{AE}$ =16cm, BE =6cm and CE =8cm. Find the length DE



12. (a) Given that $f(x) = x^2 - 5x + 6$. Find the following:

(i) Line of symmetry

(ii) Maximum or minimum value

(iii) Turning point

(iv) Intercepts

(v) Express $fx = x^2 - 5x + 6\ in\ the\ form\ of\ a(x+b)^2 + c$

13. (a) Find the image of the point (4, 2) under a reflection in the line $-x+y=0$ followed by a rotation through 90° clockwise direction about the origin.

(b) Find the image of point (3, 4) under a reflection on the $x-$ axis ($y=0$) followed by a translation, T(−4, 3).

14. (a)Given the step function $f(x) = \begin{cases} x & if\ 0 \le x < 2 \\ 3 & if\ 2 < x \le 4 \\ -4 & if\ 4 < x \le 6 \end{cases}$

i. Sketch the graph of $f(x)$

ii. State the domain and range of $f(x)$

iii. Find $f(\pi)$

(b). Given that $R=\{(x,y) : x<y, x+y>0 \text{ and } y\leq 2\}$. Draw the graph of the relation and state the domain and range of R.